Master Thesis Proposal

AI-Driven Resource Demand and Allocation Recommendation for CAPEX Portfolio Management

Author:	Ramon Bischofberger
Matriculation No.:	*<Matriculation Number>*
Supervisor:	Dr. Emanuele Laurenzi
Second Supervisor:	*<Second Supervisor>*

FHNW University of Applied Sciences and Arts Northwestern Switzerland
School of Business

Master of Science in Business Information Systems

March 23, 2026

Statement of Authenticity

I confirm that this master thesis research was performed autonomously by myself using only the sources, aids and assistance stated in the report, and that quotes are readily identifiable as such.

Date, First Name and Name

Abstract

Placeholder — write abstract here.

Contents

Statement of Authenticity **i**

Abstract **ii**

1 Introduction **1**

1.1 Problem Statement 1
1.2 Thesis Statement, Research Questions and Objectives 2
1.3 Delineations and Limitations 3
1.4 Assumptions 4
1.5 Significance 5
1.6 Chapter Map 5

2 Literature Review **7**

2.1 Search Strategy 7
2.2 CAPEX Portfolio Management and Strategic Alignment 7
2.3 Project Preemption and Suspension Mechanisms 9
2.4 Resource-Constrained Project Scheduling with Skill Considerations . . 10
2.5 Prescriptive Analytics and AI-Driven Decision Support 11
2.6 Multi-Criteria Decision Making for Project Prioritisation 12
2.7 Workforce and Skill Demand Forecasting 12
2.8 Hybrid AI: Symbolic, Subsymbolic, and Their Integration 13
2.9 Research Gap and Positioning 15

3 Research Method **17**

3.1 Philosophy 17
3.2 Approach 18
3.3 Methodological Choice 18
3.4 Strategy 18
3.5 Time Horizon 18
3.6 Data Collection 18
3.7 Data Analysis 19
3.8 Methodological Framework: AI-4-SME Design–Build–Run 19

3.9 Work Plan, Milestones, and Risks 20
3.10 Ethical Considerations 21

4 Conclusion **22**

Abbreviations **27**

List of Figures **28**

List of Tables **29**

A Literature Screening Sheet **30**

B Supervisor Documents **31**

Chapter 1

Introduction

Organizations executing large-scale Capital Expenditure (CAPEX) portfolios face a fundamental tension: multiple concurrent projects compete for the same finite pool of specialized engineering skills, yet conventional planning tools offer no objective mechanism for resolving these conflicts in a strategically coherent way. When resource shortfalls arise — as they inevitably do — portfolio managers must decide which projects continue uninterrupted and which are slowed down or paused. These decisions are today made largely through negotiation, intuition, and political influence rather than through systematic, data-driven analysis.

The consequence is a persistent misalignment between strategic portfolio priorities and operational resource deployment. High-priority projects stall because critical specialists are committed elsewhere; lower-priority work consumes capacity that cannot be redirected in time. An AI-driven prescriptive framework that can objectively recommend which projects to suspend — and when to resume them — would allow organizations to actively govern this trade-off rather than merely react to it.

1.1 Problem Statement

Current resource management in CAPEX portfolio environments is predominantly reactive and subjective. Existing tools — spreadsheets, generic ERP modules, and project portfolio management (PPM) software — can visualize resource conflicts after they materialize, but they cannot prescribe which projects to pause or resume in order to maximize strategic value under binding resource constraints. As a result:

- Suspension and resumption decisions are made ad hoc, driven by seniority or urgency rather than strategic portfolio alignment.
- Specialized engineering skills are allocated to whichever project escalates first, not to whichever project delivers the highest strategic return.

- There is no systematic mechanism to evaluate the downstream impact of suspending a project — on cost, schedule, risk, and overall portfolio value.

This gap is particularly acute in capital-intensive industries (energy, infrastructure, manufacturing) where CAPEX programs span multiple years, involve dozens of concurrent projects, and depend on a narrow set of scarce specialist roles (e.g., systems integrators, project controls engineers, domain architects). The inability to objectively arbitrate competing resource claims causes schedule slippage, budget overruns, and erosion of strategic intent.

Advances in prescriptive AI — combining optimization, simulation, and priority-scoring models — offer a principled foundation for automating these trade-off decisions (Gartner, 2024). The AI-4-SME Framework (Peter et al., 2025), developed at FHNW School of Business, provides a structured methodology for designing and deploying such AI solutions in enterprise contexts.

1.2 Thesis Statement, Research Questions and Objectives

Thesis Statement

This thesis investigates how an AI-driven prescriptive framework can be designed to support strategic resource governance in CAPEX portfolio management. The core mechanism under study is the generation of objective suspension-and-resumption recommendations: the system temporarily pauses lower-priority projects to free specialized engineering skills for projects of higher strategic importance, thereby preserving strategic alignment under binding resource constraints.

The thesis statement is:

> *The implementation of an AI-driven prescriptive framework secures the strategic alignment of a CAPEX portfolio under resource constraints by generating objective 'suspension-and-resumption' recommendations. This process optimizes project execution by temporarily pausing lower-priority tasks to release specialized engineering skills for projects of higher strategic importance.*

Research Questions

Main Research Question:

How can an AI-supported prescriptive framework enhance the strategic alignment of a CAPEX portfolio by automating project de-prioritization recommendations during specialized skill shortages?

Sub-Questions:

SQ1 How can the temporal demand for specialized technical skill pools be accurately modeled across a portfolio of 300+ ongoing projects?

SQ2 How can a dynamic weighting framework (Multi-Criteria Decision Making) be designed to integrate diverse and evolving importance indicators into an automated de-prioritization logic?

SQ3 What is the impact of systematic project suspension and resumption on overall resource availability and the successful execution of high-priority initiatives?

Research Objectives

1. Develop a strategic priority scoring model that quantifies the relative importance of projects within a CAPEX portfolio across multiple dimensions (financial, strategic, risk, time-criticality).
2. Design a prescriptive AI framework that combines priority scoring with resource constraint modeling to generate suspension-and-resumption recommendations.
3. Implement a prototype of the framework and validate its recommendations against expert judgment and simulated portfolio scenarios.
4. Derive generalizable design principles for AI-driven prescriptive systems in resource-constrained portfolio management contexts.

1.3 Delineations and Limitations

Delineations:

- The research focuses on CAPEX portfolio management in medium-to-large organizations; micro-enterprises and purely operational (OPEX) contexts are excluded.
- The prescriptive framework addresses *suspension-and-resumption* decisions at the project level; intra-project task scheduling (CPM/PERT) is outside scope.
- The resource constraints considered are *specialized human skills*; financial, equipment, and infrastructure constraints are not modeled.

- Empirical validation uses anonymized data from a partner organization or synthetic portfolio scenarios; multi-industry generalization is a direction for future work.

Limitations:

- The quality of recommendations depends on the accuracy and completeness of project priority data, which may be politically sensitive or inconsistently defined in practice.
- The study covers a defined research period; long-term organizational adoption and behavioral change are not assessed.
- The framework optimizes for strategic alignment as modeled; unmeasured organizational factors (politics, client commitments) may override recommendations in practice.

1.4 Assumptions

The following assumptions are taken as true for the purpose of this study:

- Organizations can define and quantify the strategic importance of their CAPEX projects, even if the scoring model requires elicitation and calibration.
- Project suspension and resumption are organizationally feasible actions — i.e., there exist no absolute contractual or regulatory prohibitions preventing temporary pauses.
- Specialized engineering skills are the binding resource constraint; other resources (budget, equipment) are assumed not to be simultaneously binding.
- Stakeholders and subject-matter experts who provide priority scores and validation feedback respond honestly and to the best of their knowledge.
- The AI-4-SME Framework (Peter et al., 2025), developed at FHNW, provides a methodologically sound structure for the Design, Build, and Run phases of this research.
- The regulatory environment (GDPR, EU AI Act) remains stable over the research period.

1.5 Significance

Theoretical significance: This study contributes to the body of knowledge in Business Information Systems by introducing a prescriptive AI approach to the problem of strategic resource governance in CAPEX portfolios. It extends existing work on resource-constrained project scheduling by incorporating strategic priority scoring as a first-class input to optimization, and by framing suspension-and-resumption as a deliberate governance mechanism rather than an exceptional response. The research also advances design science knowledge on how AI prescriptive systems can be constructed to produce explainable, actionable recommendations in high-stakes organizational settings.

Practical significance: Portfolio managers in capital-intensive organizations gain a systematic, objective basis for making suspension-and-resumption decisions — reducing reliance on ad hoc negotiation and political influence. The framework enables proactive reallocation of scarce engineering talent to projects where it creates the most strategic value, directly reducing schedule slippage, budget overruns, and misalignment between investment intent and execution reality. Integration with existing ERP and PPM systems provides a pathway to operational adoption.

1.6 Chapter Map

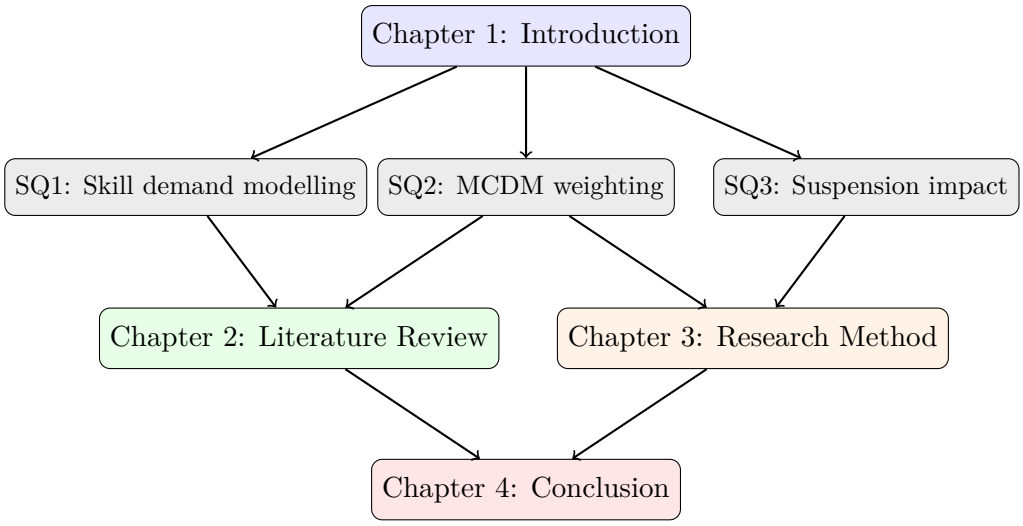
Chapter 1 — Introduction: Introduces the research problem, background, thesis statement, research questions and objectives, delineations and limitations, assumptions, and significance of the study.

Chapter 2 — Literature Review: Reviews existing work on CAPEX portfolio management, resource-constrained project scheduling, strategic priority scoring, and prescriptive AI. Identifies the research gap that motivates this thesis.

Chapter 3 — Research Method: Describes the research design, philosophical stance, methodological choices, data collection and analysis strategy, and work plan including milestones, risks, and dependencies.

Chapter 4 — Conclusion: Summarizes anticipated theoretical and practical contributions and outlines directions for future research.

The graphical chapter map below illustrates the logical flow from the research questions to the thesis chapters:


Chapter 1: Introduction
SQ1: Skill demand modelling
SQ2: MCDM weighting
SQ3: Suspension impact
Chapter 2: Literature Review
Chapter 3: Research Method
Chapter 4: Conclusion

Chapter 2

Literature Review

The literature review maps the body of knowledge relevant to the research problem and systematically identifies the gap that motivates this thesis. Following a structured search strategy, the review synthesises seven thematic clusters: (A) CAPEX portfolio management and strategic alignment, (B) project preemption and suspension mechanisms, (C) resource-constrained project scheduling with skill considerations, (D) prescriptive analytics and AI-driven decision support, (E) multi-criteria decision making for project prioritisation, (F) workforce and skill demand forecasting, and (G) hybrid AI combining symbolic and subsymbolic methods. The chapter closes with a cross-cluster gap analysis that directly motivates the thesis research questions.

2.1 Search Strategy

The systematic search was conducted on **Scopus** as the primary academic database, supplemented by targeted searches on IEEE Xplore, ACM Digital Library, and INFORMS PubsOnline for conference and journal papers not fully indexed at the time of search. The search applied Boolean keyword combinations across seven thematic clusters, as summarised in Table 2.1. A two-stage screening procedure was applied: title and abstract screening followed by full-text relevance assessment. Only papers rated Medium or High relevance against the research questions were retained for synthesis.

2.2 CAPEX Portfolio Management and Strategic Alignment

Project portfolio management (PPM) has evolved from a purely financial discipline into a strategic governance function concerned with selecting, prioritising, and terminating projects in ways that maximise organisational value (Hansen and Svejvig, 2022). In a comprehensive bibliometric analysis spanning seven decades of PPM research, Hansen

Table 2.1: Keyword clusters used in the systematic Scopus search

Cluster	Search Terms
A — Portfolio	`CAPEX portfolio management`, `capital expenditure portfolio`, `project portfolio strategic alignment`, `portfolio value optimisation`
B — Preemption	`project preemption`, `preemptive RCPSP`, `activity splitting`, `project suspension resumption`
C — RCPSP	`resource-constrained project scheduling`, `multi-skill scheduling`, `multi-project scheduling`, `skill-based resource allocation`
D — Prescriptive	`prescriptive analytics`, `decision support resource allocation`, `from predictive to prescriptive`
E — MCDM	`MCDM project prioritisation`, `AHP TOPSIS portfolio`, `dynamic weighting strategic alignment`
F — Forecasting	`workforce skill demand forecasting`, `temporal skill modelling`, `labour demand machine learning`
G — Hybrid AI	`hybrid AI knowledge graph ontology`, `neuro-symbolic AI decision support`, `ESCO skill ontology`

and Svejvig (2022) identify resource allocation under competing constraints and the dynamic re-alignment of portfolios to shifting strategic objectives as the two most persistently unresolved challenges in the field. This observation directly contextualises the present research: in CAPEX-intensive environments, resource conflicts among concurrent projects are structurally unavoidable, yet the literature offers no established automated mechanism for resolving them in a strategically consistent manner.

Hansen and Svejvig (2023) extend this line of inquiry by deriving seventeen normative PPM principles organised around four semantic dimensions — portfolio value, strategic fit, balance, and adaptive capacity. Their work provides a principled basis for evaluating automated suspension-and-resumption recommendations: such recommendations should demonstrably improve strategic fit and portfolio balance, not merely minimise makespan or resource cost.

At the level of individual portfolio optimisation models, Chiang and Nunez (2013) propose an integer programming formulation for IT project portfolios that integrates a strategic alignment score alongside expected benefits, development costs, and cross-project synergies. Their model demonstrates that explicit strategic scoring, rather than implicit proxy measures such as financial return, is essential for producing portfolios that genuinely reflect organisational priorities. The same principle applies to CAPEX portfolios, where strategic objectives — energy transition targets, regulatory compliance milestones, market positioning — are often only loosely coupled with financial metrics. More recently, Rodriguez-Garcia et al. (2025) extend this direction by combining machine learning with mathematical programming to simultaneously optimise for strategic factors and synergies; their empirical results show that strategy-aware portfolio

selection consistently outperforms purely financial approaches, even under uncertainty.

The growing complexity of CAPEX programmes — driven by regulatory pressures, talent scarcity, and technological change — is reflected in the most recent contribution by Rodrigues Coelho et al. (2025), who apply Structural Equation Modelling and Necessary Condition Analysis to demonstrate that PPM is a necessary but not sufficient mediator between organisational flexibility and value creation. The implication for AI-driven portfolio governance is significant: strategic alignment must be embedded not only as an objective function but as a hard constraint that any automated recommendation must respect.

2.3 Project Preemption and Suspension Mechanisms

The concept of temporarily suspending an ongoing project activity to release a resource for a higher-priority demand has a formal treatment in the operations research literature under the label of *preemption*. The foundational study by Ballestín et al. (2008) demonstrates that allowing even a single preemptive interruption within a resource-constrained project scheduling problem (RCPSP) yields statistically significant makespan reductions relative to purely non-preemptive schedules. This result establishes preemption as an operationally meaningful mechanism, not merely a theoretical relaxation.

Van Peteghem and Vanhoucke (2010) extend this analysis to multi-mode RCPSP, showing that a genetic algorithm capable of selecting between preemptive and non-preemptive execution modes consistently produces superior schedules when resource availability is uneven across the planning horizon. Their work is particularly relevant to CAPEX portfolios, where the demand profile for specialised engineering skills fluctuates significantly across project lifecycle phases.

The most practically relevant model for the present thesis is provided by Hatami-Moghaddam et al. (2024), who introduce a robust multi-skill, multi-mode RCPSP formulation with partial preemption. In their model, an activity may be suspended precisely when a required skill set becomes unavailable within a specified time window, and resumed when the skill pool is replenished. This mechanism maps directly onto the suspension-and-resumption logic proposed in the thesis, with specialised engineering skill shortages serving as the trigger condition.

Vanhoucke et al. (2025) contribute a bi-criteria analysis examining the trade-off between makespan reduction and the number of interruptions induced by preemption in problems with time-varying resources. Their finding that optimal suspension frequency is sensitive to resource availability patterns suggests that a data-driven, dynamic policy — as proposed by the prescriptive AI framework in this thesis — is preferable to a static threshold-based rule.

Collectively, this body of work establishes that project suspension is a formally

sound and empirically effective mechanism for releasing constrained resources, yet existing models operate on single projects or small portfolios and employ rule-based or combinatorial triggers rather than strategic priority scoring. This is a first dimension of the research gap.

2.4 Resource-Constrained Project Scheduling with Skill Considerations

The resource-constrained project scheduling problem (RCPSP) is one of the most extensively studied problems in operations research. Artigues et al. (2025) trace fifty years of RCPSP research, documenting the progression from exact branch-and-bound methods to metaheuristics, and from single-project formulations to the multi-skill, multi-project variants most relevant to contemporary CAPEX portfolio management. Hartmann and Briskorn (2022) provide a complementary decade-level update, identifying skill-based and multi-project RCPSP as the fastest-growing research sub-areas.

For portfolios of the scale addressed in this thesis (300+ concurrent projects), Gómez Sánchez et al. (2023) demonstrate that shared resource pools across concurrent projects are the primary source of scheduling conflicts and that the resulting *resource-constrained multi-project scheduling problem* (RCMPSP) is computationally intractable for exact methods at real-world scale. This motivates the use of heuristic and data-driven approaches.

The modelling of human skill heterogeneity within RCPSP has received increasing attention. Snauwaert and Vanhoucke (2021) introduce a distinction between *skill breadth* (the number of distinct skills an individual possesses) and *skill depth* (proficiency level per skill), demonstrating that both dimensions independently affect assignment quality in a genetic algorithm framework. Snauwaert and Vanhoucke (2023) formalise a classification scheme for multi-skilled RCPSP (MSRCPSP) and release standardised benchmark instances, providing the empirical infrastructure needed for rigorous model evaluation. Their most recent contribution (Snauwaert and Vanhoucke, 2025) introduces six MSRCPSP variants with hierarchical skill structures — where junior skills are subsumed by senior ones — directly mirroring the tiered engineering competency pools typical of large CAPEX programmes.

The industrial feasibility of solving large-scale MSRCMPSP instances is demonstrated by Torba et al. (2024), who address a real-world scheduling problem at SNCF heavy maintenance involving thousands of activities, multiple skill types, and weighted tardiness objectives. Their memetic algorithm solution approach is the closest real-world analogue to the computational context of the present thesis. Similarly, Haroune et al. (2023) address multi-project assignment under shared employee skill pools with efficiency

levels, minimising both weighted tardiness and goal deviations — a formulation that naturally accommodates project priority weights, linking Cluster C to the MCDM considerations of Cluster E.

2.5 Prescriptive Analytics and AI-Driven Decision Support

Prescriptive analytics denotes the class of analytical methods that go beyond description and prediction to generate actionable recommendations — that is, explicit decisions or interventions designed to achieve a specified objective under identified constraints (Lepenioti et al., 2020). In their systematic review of 56 key papers, Lepenioti et al. (2020) identify resource allocation as the primary application domain of prescriptive methods and distinguish three core technical approaches: optimisation, simulation, and heuristic rule learning. Frazzetto et al. (2019) complement this picture from a data management perspective, identifying constraint satisfaction, optimisation, and simulation as the architectural pillars of any prescriptive analytics system — a decomposition that maps directly onto the design components of the proposed framework.

The theoretical foundation for embedding prediction within optimisation is provided by Bertsimas and Kallus (2020), who formalise the transition from predictive to prescriptive analytics and introduce the *coefficient of prescriptiveness* as a performance measure capturing the value added by incorporating predictions into decisions. Their framework is particularly relevant to the thesis's analytics chain: temporal skill demand forecasts (the predictive layer) are the principal input to the suspension-and-resumption optimisation (the prescriptive layer).

Wissuchek and Zschech (2025) offer the most recent and comprehensive systems-theoretic treatment of prescriptive analytics systems (PAS), identifying 23 architectural components including data pipeline, intervention model, feedback mechanism, and explainability layer. Their analysis reveals that few existing PAS implementations address multi-objective portfolio decisions with dynamic constraint sets — precisely the configuration required for CAPEX resource governance. Mišić and Perakis (2020) further contextualise prescriptive methods within the broader operations management literature, noting that their comparative advantage over predictive methods is most pronounced when decision windows are short and resource constraints are binding, which is consistently the case in CAPEX portfolio management.

2.6 Multi-Criteria Decision Making for Project Prioritisation

Project prioritisation in the context of competing resource constraints is inherently a multi-criteria problem: strategic importance, financial return, regulatory urgency, stakeholder commitment, and execution risk are incommensurable attributes that must be integrated into a single ranking or weighting scheme. Kandakoglu et al. (2024) conduct a systematic literature review of MCDM methods applied to project portfolio selection, covering the Analytic Hierarchy Process (AHP), PROMETHEE, TOPSIS, and Data Envelopment Analysis (DEA). A central finding of their review is that *dynamic re-prioritisation under constraint changes* is an open research problem: existing MCDM applications to PPM are almost uniformly static, producing a priority ordering at a single point in time rather than continuously updating weights as resource availability and strategic context evolve. This constitutes a second dimension of the research gap.

Chatterjee et al. (2018) address the treatment of linguistic uncertainty in expert prioritisation judgements through fuzzy AHP. Their contribution is methodologically important for the present thesis: criteria weights assigned by portfolio managers are inherently imprecise and may change over time, requiring a weighting framework that accommodates vagueness without reducing to arbitrary scores. The dynamic extension of this approach — where fuzzy weights are recalibrated in response to changing resource conditions and project progress — forms the conceptual basis of Sub-Question 2.

The application of MCDM to capital-intensive industrial portfolios is examined by Al-Hashimi et al. (2017), who compare the output coherence of AHP, PROMETHEE, and TOPSIS applied to oilfield CAPEX portfolio prioritisation. Their key finding — that method output coherence is high when criteria weights are well-calibrated but diverges under expert disagreement — underscores the need for systematic weight elicitation and sensitivity analysis in any automated de-prioritisation system.

2.7 Workforce and Skill Demand Forecasting

Accurately modelling the temporal demand for specialised skill pools is a prerequisite for any anticipatory suspension-and-resumption mechanism. Without a reliable forecast of when and where skill shortages will materialise, suspension recommendations can only be generated reactively — after conflicts have already disrupted project execution. Safarishahrbijari (2018) provide a systematic review of workforce forecasting models, covering time series methods, Markov chain approaches, system dynamics, and discrete event simulation. Their taxonomy reveals a consistent pattern: models that integrate demand-side dynamics (project workload profiles) with supply-side constraints (skill availability, attrition, hiring lead times) outperform demand-only models in medium-to-

long forecast horizons. For a 300+ project portfolio with overlapping lifecycle phases, such integrated modelling is essential.

At the intersection of deep learning and skill demand forecasting, Macedo et al. (2022) apply recurrent neural networks (LSTM) to a decade of monthly skill demand observations, demonstrating that multivariate models capturing inter-skill correlations outperform univariate baselines in multi-step forecasting tasks. Their work provides empirical evidence that the temporal skill demand patterns required by Sub-Question 1 are learnable from historical project staffing data, subject to sufficient data quality and volume.

Temporal Fusion Transformers (Lim et al., 2021) represent the current state of the art for multi-horizon time series forecasting with mixed-frequency inputs and static covariates. Their architecture — combining multi-head attention for long-range dependencies with interpretable variable selection — is a natural candidate for skill demand forecasting in CAPEX portfolios, where project metadata (type, phase, contractual milestones) constitutes structured static covariates that modulate temporal demand patterns.

2.8 Hybrid AI: Symbolic, Subsymbolic, and Their Integration

A recurring limitation of purely data-driven approaches in enterprise decision support is their dependence on the statistical regularity of historical data. In domains characterised by low data volume, high structural complexity, or rapidly changing operational contexts — all of which apply to CAPEX portfolio management — subsymbolic models alone are insufficient. The paradigm of *hybrid AI* addresses this limitation by combining the complementary strengths of symbolic and subsymbolic methods (Garcez and Lamb, 2023).

Symbolic AI encompasses methods that operate on explicit, human-interpretable representations of knowledge: ontologies, rule systems, logic programs, and knowledge graphs. Symbolic methods are inherently interpretable, capable of encoding domain constraints formally, and effective in low-data regimes. Their weakness is brittleness: they cannot generalise beyond explicitly encoded knowledge and fail gracefully when assumptions are violated.

Subsymbolic AI encompasses statistical learning methods — including classical machine learning, deep learning, and generative AI — that extract patterns from data without explicit symbolic encoding. Subsymbolic models are powerful pattern recognisers that generalise to unseen inputs, but they lack interpretability, require large training sets, and cannot natively incorporate hard domain constraints.

Hybrid AI combines both paradigms, typically by using structured symbolic knowledge to constrain, enrich, or initialise subsymbolic learning processes. Gartner identifies hybrid AI as an emerging high-value technology trend in enterprise AI adoption, highlighting its particular relevance for domains where domain expertise must be preserved while adapting to changing data distributions (Gartner, 2024). Garcez and Lamb (2023) formalise this integration as *neurosymbolic AI*, demonstrating that combining neural learning with symbolic reasoning yields systems that are simultaneously more accurate and more interpretable than either component in isolation.

In the context of this thesis, the hybrid AI architecture serves three distinct roles. First, a symbolic layer encodes domain knowledge about skill taxonomies, project types, and organisational constraints in a structured, machine-processable form. Second, a subsymbolic layer learns temporal demand patterns from historical project staffing data, with the symbolic layer providing structured features and constraint boundaries. Third, a prescriptive layer integrates both outputs into suspension-and-resumption recommendations that are both data-driven and domain-knowledge-compliant.

The ESCO Ontology provides the symbolic backbone for skill representation in this framework. Developed by the European Commission, ESCO (European Skills, Competences, Qualifications and Occupations) is a multilingual, hierarchical taxonomy of over 13,900 skills and competences formally linked to occupations and qualifications through a structured ontological hierarchy (European Commission, 2014). ESCO's formal structure — skills organised into pillar hierarchies with defined relationships (`skos:broader`, `skos:narrower`, `skos:related`) — makes it directly suitable as a semantic backbone for skill-demand modelling in CAPEX portfolios. Specifically, ESCO enables consistent skill labelling across heterogeneous project data sources, ontological inference of skill substitutability (i.e., which alternative skills can fulfil a demand when the primary skill is unavailable), and semantic enrichment of subsymbolic demand forecasting models through ontology embeddings.

The AI-4-SME Framework (Peter et al., 2025), developed at FHNW School of Business, provides the overarching methodological structure for translating the hybrid AI architecture into an implementable solution. The framework's Design–Build–Run methodology distinguishes Knowledge-Intensive Activities (KIAs) — which require contextual judgement, problem solving, and expert knowledge — from Data-Intensive Activities (DIAs) — which require high-volume data processing and statistical learning. In the CAPEX resource management context, demand forecasting and strategic priority scoring are KIAs; allocation optimisation and constraint propagation are DIAs. The framework's hybrid build path, combining knowledge-based systems (Neo4j, GraphDB, Stardog) with data-based ML pipelines (TensorFlow, PyTorch), maps directly onto the proposed framework architecture.

2.9 Research Gap and Positioning

The synthesis of the seven clusters reveals a coherent set of gaps at the intersection of the reviewed domains. Table 2.2 summarises the state of knowledge per dimension and identifies the contribution of the present thesis.

Table 2.2: Research gap summary across literature clusters

Dimension	**State of the Art**	**Gap / Thesis Contribution**
Strategic alignment in PPM	Static portfolio selection models with strategic scoring (Chiang and Nunez, 2013; Rodriguez-Garcia et al., 2025)	No dynamic, AI-driven realignment triggered by real-time skill shortages
Preemptive project scheduling	Rule-based and combinatorial preemption models for single or small portfolios (Ballestín et al., 2008; Hatami-Moghaddam et al., 2024)	No prescriptive AI layer translating portfolio-level strategic priorities into suspension triggers
Multi-skill RCPSP at scale	Heuristic and metaheuristic methods for industrial-scale instances (Torba et al., 2024; Snauwaert and Vanhoucke, 2025)	No integration with strategic MCDM weighting or AI-driven priority scoring
Prescriptive analytics	Theoretical frameworks and isolated domain applications (Lepenioti et al., 2020; Wissuchek and Zschech, 2025)	No PAS implementation for multi-project CAPEX suspension-and-resumption decisions
MCDM for project prioritisation	Static, point-in-time ranking with expert-elicited weights (Kandakoglu et al., 2024; Chatterjee et al., 2018)	No dynamic, continuously updated MCDM weighting framework responsive to resource availability and project progress
Skill demand forecasting	Temporal models for labour market or HR demand (Safarishahrbijari, 2018; Macedo et al., 2022)	No application to project-level, portfolio-wide skill pool demand across 300+ concurrent CAPEX projects
Hybrid AI in enterprise decision support	Neurosymbolic frameworks and ESCO-based skill extraction (Garcez and Lamb, 2023; Gartner, 2024)	No hybrid AI system combining ESCO ontology, temporal demand forecasting, and prescriptive suspension optimisation in a unified CAPEX framework

The overarching research gap can be stated as follows: *no existing work unifies (1) strategic priority scoring via dynamic MCDM, (2) AI-driven temporal modelling of specialised skill pool demand, and (3) prescriptive suspension-and-resumption optimisation into a single, operationally deployable framework for CAPEX portfolio management.* Each of the three components exists in isolation within its respective literature cluster;

their integration, guided by the AI-4-SME framework and grounded in ESCO-based hybrid AI architecture, constitutes the novel contribution of this thesis.

Chapter 3

Research Method

The purpose of this chapter is to describe the research design and convince the reader that it is appropriate to answer the research questions and validate the thesis statement. The chapter follows the structure of the Research Onion, presenting only the choices made for this study.

Table 3.1: Research design choices

Layer	**Choice**
Philosophy (only 1)	Pragmatism
Approach (only 1)	Abductive
Methodological Choice (only 1)	Mono — Quantitative (artifact + evaluation)
Strategy (only 1)	Design Science Research (DSR)
Time Horizon (only 1)	Cross-sectional
Data Collection (1 or more)	Historical staffing data; expert interviews
Data Analysis (1 or more)	ML model training & evaluation; thematic coding

3.1 Philosophy

The philosophical stance for this research is **pragmatism**. Pragmatism is appropriate because the primary aim is to develop a working artifact (the hybrid AI recommendation engine) that solves a concrete organizational problem. Truth is assessed by practical outcomes — whether the system demonstrably improves bottleneck anticipation — rather than by adherence to a single ontological or epistemological paradigm. This stance allows combining qualitative domain knowledge elicitation (symbolic AI design) with quantitative model evaluation (subsymbolic AI training and testing).

3.2 Approach

An **abductive** approach is followed: starting from observed anomalies in CAPEX resource planning practice, the research moves iteratively between theory (ontologies, ML forecasting literature) and empirical data (project staffing records) to build and refine the best explanatory model. This differs from pure induction (no prior theory) and pure deduction (testing a fixed hypothesis without iteration), matching the iterative nature of Design Science Research cycles.

3.3 Methodological Choice

The methodological choice is **mono-method with a quantitative emphasis**: the core research activity is the design, implementation, and quantitative evaluation of an IT artifact. Qualitative elements (expert interviews for domain knowledge acquisition and requirements validation) are used instrumentally within the DSR process and do not constitute a separate research strand.

3.4 Strategy

The strategy is **Design Science Research (DSR)** (Hevner et al., 2004), which is appropriate for the development and evaluation of IT artifacts in applied organizational contexts. The primary artifact is the hybrid AI recommendation engine; the research contribution lies in both the artifact itself and the generalizable design knowledge extracted from its development and evaluation. DSR is structured around three cycles: the *Relevance Cycle* (problem and requirements from practice), the *Design Cycle* (build and evaluate), and the *Rigor Cycle* (grounding in existing knowledge base). This maps directly onto the AI-4-SME Design–Build–Run phases (Peter et al., 2025).

3.5 Time Horizon

The time horizon is **cross-sectional**: data collection, model training, and evaluation are performed within a defined period (approx. January 2026 to September 2026). This restricts the ability to capture long-term adoption trends and reduces generalizability of performance results; this is explicitly acknowledged as a limitation (see Section 1.3).

3.6 Data Collection

Two data collection methods are used:

1. **Historical project staffing data:** Anonymized records from a partner organization (or synthetic benchmarks generated from industry data) covering project metadata, role assignments, skill tags, and actuals vs. planned allocations. This data underpins the quantitative demand forecasting models.

2. **Expert interviews / workshops:** Semi-structured interviews with portfolio managers and resource planners to (a) elicit domain knowledge for ontology design, (b) validate the recommendation engine output against expert judgment, and (c) assess perceived decision support quality. The following guiding questions structure the interviews:

Table 3.2: Interview guiding questions

Question	**Purpose**
How do you currently identify upcoming resource bottlenecks in the portfolio?	Understand baseline process and pain points
What data sources do you rely on for staffing decisions, and how reliable are they?	Identify available data assets and quality issues
Which skill categories are most difficult to plan for and why?	Prioritize ESCO coverage requirements
What would an ideal recommendation look like, and what would make you trust it?	Define output requirements and explainability needs
What constraints (budget, regulation, HR policy) must any recommendation respect?	Capture constraint requirements for allocation engine

3.7 Data Analysis

- **Quantitative:** ML model training and evaluation using standard metrics (MAE, RMSE for forecasting; precision/recall for bottleneck classification). Comparison against a baseline (moving average / current ERP planning output).

- **Qualitative:** Thematic coding of interview transcripts to extract design requirements and validation feedback. Codes are organized around the six W-questions of the AI-4-SME task design method.

3.8 Methodological Framework: AI-4-SME Design–Build–Run

The AI-4-SME Framework (Peter et al., 2025) structures the research across three phases:

Phase 1 — Design

- **Company level:** Identify CAPEX resource management as the strategic AI application domain; assess tasks for AI suitability using the feasibility-vs-impact matrix.
- **Process level:** Map the CAPEX project lifecycle; identify KIAs (skill-demand forecasting, bottleneck detection) and DIAs (allocation optimization, capacity computation).
- **Task level:** Apply the six W-questions to define the point-of-view for each AI task; generate solution ideas and select the hybrid path based on effort-vs-benefit.

Phase 2 — Build (Hybrid AI)

1. **Knowledge-based stream:** Acquire and represent domain knowledge using ESCO as the skill ontology backbone; implement a knowledge graph (Neo4j or GraphDB).
2. **Data-based stream:** Prepare historical staffing data; train and evaluate ML models (gradient boosting, LSTM, Temporal Fusion Transformer) for temporal skill-demand forecasting using knowledge graph embeddings as features.
3. **Hybrid integration:** Combine both streams into a unified recommendation engine delivering ontology-grounded, data-driven allocation recommendations.

Phase 3 — Run

- **Governance:** Evaluate GDPR and EU AI Act compliance (transparency, explainability, risk classification).
- **IT integration:** Define API interface for ERP and PPM system integration.
- **Human-AI collaboration:** Design the recommendation interface to support — not replace — portfolio manager decision-making.

3.9 Work Plan, Milestones, and Risks

Key risks and dependencies:

- *Data availability risk:* If real organizational data cannot be obtained, synthetic data will be generated from published industry benchmarks. (Mitigation: early agreement with partner organization; synthetic fallback planned.)

Table 3.3: Work packages and milestones

#	Task / Work Package	Output	Duration	Milestone
1	Literature review & gap analysis	Screened corpus, gap statement	4 weeks	M1: Literature finalized
2	Domain knowledge elicitation (interviews)	Interview data, ontology requirements	3 weeks	M2: Requirements validated
3	ESCO knowledge graph design & build	Knowledge graph prototype	4 weeks	M3: KG operational
4	Data preparation & feature engineering	ML-ready dataset	3 weeks	
5	ML model development & training	Trained forecasting models	5 weeks	M4: Models benchmarked
6	Hybrid integration & recommendation engine	Integrated prototype	4 weeks	M5: Prototype complete
7	Evaluation & expert validation	Evaluation results	3 weeks	M6: Evaluation done
8	Thesis writing & revision	Final thesis document	6 weeks	M7: Thesis submitted

- *Ontology coverage risk:* ESCO may not cover highly specialized CAPEX roles. (Mitigation: manual extension nodes linked to ESCO via `skos:broader`.)
- *Integration complexity:* Combining KG embeddings with ML pipelines requires tight interface design. (Mitigation: modular architecture with defined APIs between components.)
- *Expert availability:* Interview scheduling with senior portfolio managers may be delayed. (Mitigation: recruit multiple candidates early; parallelize with data preparation.)

3.10 Ethical Considerations

If real organizational data is used, appropriate data anonymization and confidentiality agreements will be established in advance. The recommendation engine is designed to support — not replace — human decision-making, with explainability mechanisms ensuring transparency of recommendations. Compliance with GDPR and the EU AI Act is built into the Run phase of the AI-4-SME methodology.

Chapter 4

Conclusion

This proposal has presented a research design for developing and evaluating a hybrid AI recommendation engine for CAPEX resource management. The central argument is that combining ESCO-grounded knowledge graphs with data-driven demand forecasting models — structured according to the AI-4-SME Design–Build–Run methodology — can substantially improve the anticipation of skill bottlenecks and the quality of allocation recommendations compared to conventional spreadsheet-based planning tools.

The four research questions address distinct aspects of this challenge: the design of the hybrid AI architecture (RQ1), the selection of appropriate symbolic knowledge structures (RQ2), the translation of detected bottlenecks into actionable recommendations (RQ3), and the empirical evaluation of the resulting system (RQ4). Taken together, their answers will validate or refute the thesis statement.

Anticipated theoretical contributions:

- Design principles for knowledge-augmented forecasting systems in resource management.
- An empirically grounded evaluation of hybrid AI architectures in an applied organizational setting.
- Refinement of the AI-4-SME framework through application to a complex, multi-stakeholder CAPEX domain.

Anticipated practical contributions:

- A prototype recommendation engine demonstrating how ESCO skill data and historical project staffing records can be combined to generate proactive workforce decisions.
- A replicable methodology for organizations seeking to implement similar systems within existing ERP/PPM infrastructure.

- Governance guidelines for deploying hybrid AI recommendation engines in compliance with GDPR and the EU AI Act.

Future research: Future work could extend the evaluation to multiple organizations to test generalizability, explore real-time integration with ERP systems, and investigate human-AI collaboration patterns in the context of portfolio manager decision-making.

Bibliography

Al-Hashimi, M., Chakrabortty, R. K., and Ryan, M. J. (2017). Analysis of MCDM methods output coherence in oil and gas portfolio prioritization. *Journal of Petroleum Exploration and Production Technology*, 7(4):1103–1116.

Artigues, C., Hartmann, S., and Vanhoucke, M. (2025). Fifty years of research on resource-constrained project scheduling explored from different perspectives. *European Journal of Operational Research*, 328(2).

Ballestín, F., Valls, V., and Quintanilla, S. (2008). Pre-emption in resource-constrained project scheduling. *European Journal of Operational Research*, 189(3):1136–1152.

Bertsimas, D. and Kallus, N. (2020). From predictive to prescriptive analytics. *Management Science*, 66(3):1025–1044.

Chatterjee, K., Hossain, S. A., and Kar, S. (2018). Prioritization of project proposals in portfolio management using fuzzy AHP. *OPSEARCH*, 55(2):478–501.

Chiang, I. R. and Nunez, M. A. (2013). Strategic alignment and value maximization for IT project portfolios. *Information Technology and Management*, 14(2):143–157.

European Commission (2014). ESCO: European skills, competences, qualifications and occupations. Publications Office of the European Union.

Frazzetto, D., Nielsen, T. D., Pedersen, T. B., and Šikšnys, L. (2019). Prescriptive analytics: A survey of emerging trends and technologies. *The VLDB Journal*, 28(4):575–595.

Garcez, A. d. and Lamb, L. C. (2023). Neurosymbolic AI: The 3rd wave. *Artificial Intelligence Review*, 56(11):12387–12406.

Gartner (2024). Gartner hype cycle for artificial intelligence. Gartner Research Report.

Gómez Sánchez, M., Lalla-Ruiz, E., Fernández Gil, A., Castro, C., and Voß, S. (2023). Resource-constrained multi-project scheduling problem: A survey. *European Journal of Operational Research*, 309(3):958–976.

Hansen, L. K. and Svejvig, P. (2022). Seven decades of project portfolio management research (1950–2019) and perspectives for the future. *Project Management Journal*, 53(4):402–423.

Hansen, L. K. and Svejvig, P. (2023). Principles in project portfolio management: Building upon what we know to prepare for the future. *Project Management Journal*, 54(5):475–489.

Haroune, M., Dhib, C., Neron, E., et al. (2023). Multi-project scheduling problem under shared multi-skill resource constraints. *TOP*.

Hartmann, S. and Briskorn, D. (2022). An updated survey of variants and extensions of the resource-constrained project scheduling problem. *European Journal of Operational Research*, 297(1):1–14.

Hatami-Moghaddam, L., Khalilzadeh, M., Shahsavari-Pour, N., and Sajadi, S. M. (2024). Developing a robust multi-skill, multi-mode resource-constrained project scheduling model with partial preemption, resource leveling, and time windows. *Mathematics*, 12(19):3129.

Hevner, A. R., March, S. T., Park, J., and Ram, S. (2004). Design science in information systems research. *MIS Quarterly*, 28(1):75–105.

Kandakoglu, M., Walther, G., and Ben Amor, S. (2024). The use of multi-criteria decision-making methods in project portfolio selection: A literature review and future research directions. *Annals of Operations Research*.

Lepenioti, K., Bousdekis, A., Apostolou, D., and Mentzas, G. (2020). Prescriptive analytics: Literature review and research challenges. *International Journal of Information Management*, 50:57–70.

Lim, B., Arik, S. O., Loeff, N., and Pfister, T. (2021). Temporal fusion transformers for interpretable multi-horizon time series forecasting. *International Journal of Forecasting*, 37(4):1748–1764.

Macedo, M. M. G., Clarke, W., Lucherini, E., Baldwin, T., Queiroz, D., de Paula, R., and Das, S. (2022). Practical skills demand forecasting via representation learning of temporal dynamics. In *Proceedings of the 2022 AAAI/ACM Conference on AI, Ethics, and Society*, pages 430–440. ACM.

Mišić, V. V. and Perakis, G. (2020). Data analytics in operations management: A review. *Manufacturing & Service Operations Management*, 22(1):158–169.

Peter, M. K., Laurenzi, E., and Hinkelmann, K. (2025). *Workshop Canvas AI-4-SME Framework: Identification and utilisation of strategic AI opportunities to strengthen a firm's competitiveness and innovation capabilities*. FHNW School of Business, Switzerland. Version 06/25. Available at https://www.ai-framework.net.

Rodrigues Coelho, F. I., Bizarrias, F. S., Rabechini, R., Martens, C. D. P., and Martens, M. L. (2025). Strategic alignment and value optimization: Unveiling the critical role of project portfolio management for a flexible environment. *Global Journal of Flexible Systems Management*.

Rodriguez-Garcia, P., Juan, A. A., Martin, J. A., Lopez-Lopez, D., and Marco, J. M. (2025). AI-driven optimization of project portfolios in corporate ecosystems with synergies and strategic factors. *Expert Systems with Applications*.

Safarishahrbijari, A. (2018). Workforce forecasting models: A systematic review. *Journal of Forecasting*, 37(7):739–753.

Snauwaert, J. and Vanhoucke, M. (2021). A new algorithm for resource-constrained project scheduling with breadth and depth of skills. *European Journal of Operational Research*, 292(1):43–58.

Snauwaert, J. and Vanhoucke, M. (2023). A classification and new benchmark instances for the multi-skilled resource-constrained project scheduling problem. *European Journal of Operational Research*, 307(1):1–19.

Snauwaert, J. and Vanhoucke, M. (2025). A solution framework for multi-skilled project scheduling problems with hierarchical skills. *Journal of Scheduling*.

Torba, R., Dauzère-Pérès, S., Yugma, C., Gallais, C., and Pouzet, J. (2024). Solving a real-life multi-skill resource-constrained multi-project scheduling problem. *Annals of Operations Research*.

Van Peteghem, V. and Vanhoucke, M. (2010). A genetic algorithm for the preemptive and non-preemptive multi-mode resource-constrained project scheduling problem. *European Journal of Operational Research*, 201(2):409–418.

Vanhoucke, M., Demeulemeester, E., and Herroelen, W. (2025). The impact of the number of preemptions in resource-constrained project scheduling problems with time-varying resources: Computational experiments and a case study. *Annals of Operations Research*.

Wissuchek, C. and Zschech, P. (2025). Prescriptive analytics systems revised: A systematic literature review from an information systems perspective. *Information Systems and e-Business Management*.

Abbreviations

AI Artificial Intelligence

API Application Programming Interface

CAPEX Capital Expenditure

CPM Critical Path Method

DIA Data-Intensive Activity

DSR Design Science Research

ERP Enterprise Resource Planning

ESCO European Skills, Competences, Qualifications and Occupations

FHNW Fachhochschule Nordwestschweiz (University of Applied Sciences and Arts Northwestern Switzerland)

GDPR General Data Protection Regulation

KG Knowledge Graph

KIA Knowledge-Intensive Activity

LSTM Long Short-Term Memory

ML Machine Learning

OPEX Operational Expenditure

PPM Project Portfolio Management

RCPSP Resource-Constrained Project Scheduling Problem

SME Small and Medium-sized Enterprise

List of Figures

List of Tables

2.1 Keyword clusters used in the systematic Scopus search 8
2.2 Research gap summary across literature clusters 15

3.1 Research design choices 17
3.2 Interview guiding questions 19
3.3 Work packages and milestones 21

A.1 Literature screening sheet columns 30

Appendix A

Literature Screening Sheet

The following columns are used in the Scopus literature screening Excel file:

Table A.1: Literature screening sheet columns

Column	**Description**
Author(s)	Author names
Year	Publication year
Title	Full paper title
Source	Journal / Conference
Relevance	Low / Medium / High — assessed against the four RQs
Finding	Key insight relevant to this research
Category	Which RQ or theme the paper primarily addresses
Notes	Additional remarks

Appendix B

Supervisor Documents

The following documents were provided by supervisor Dr. Emanuele Laurenzi and are available in the `documents/` folder:

- `03-Workshop-Canvas-KI-KMU-EN.pdf` — AI workshop canvas for SMEs (EN)
- `FHNW-KI-KMU-Methodik-Praxisleitfaden.pdf` — FHNW AI methodology guide for SMEs